ARJUNA

JEE 2026 BATCH

Chemistry

Chemical Equilibrium

Lecture No- 02

By – Rahul Dudi Sir

Topics to be covered

1. Concept of equilibrium
2. Physical equilibrium
3. Chemical equilibrium

Chalo shuru
karte hai
Physics Wallah

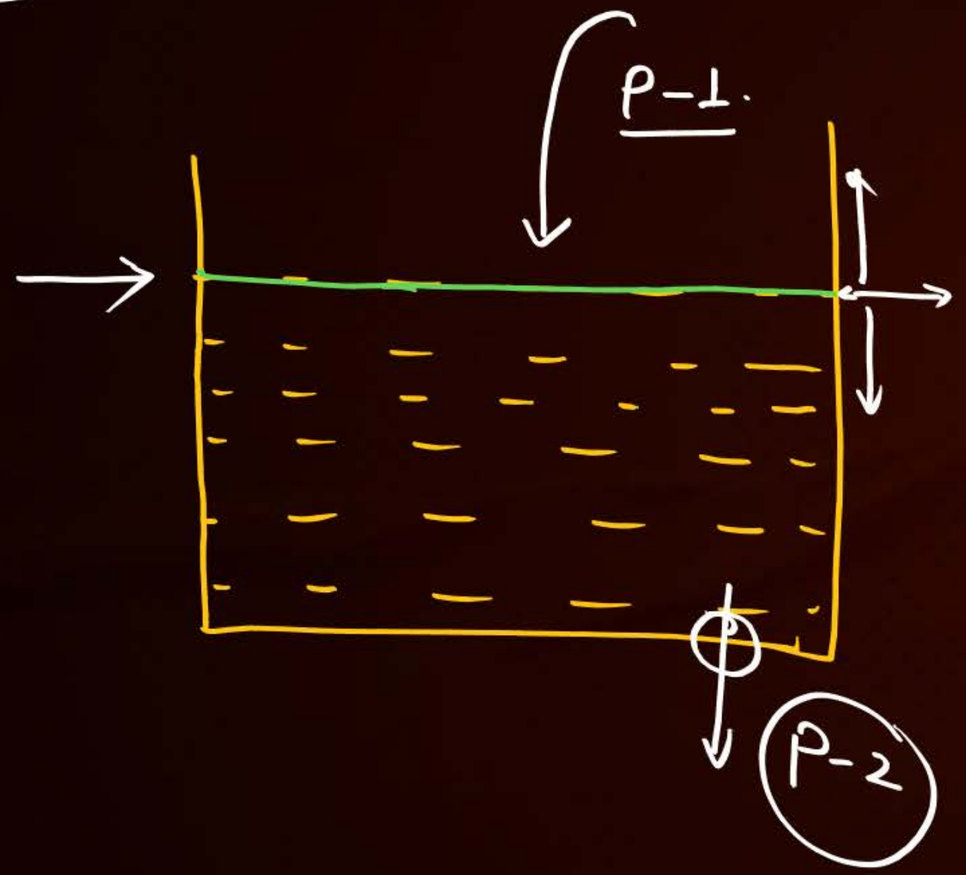
Types of Chemical Reactions

- On the basis of completion, there are two types of chemical reactions–
-

1. Reversible reactions
2. Irreversible reactions

Concept of Equilibrium

Concept of Equilibrium



P-1 and P-2 → opposite process.

Same rate.

Eq.
- → Static
- → Dynamic.

Done

Concept of Equilibrium

Done

- Equilibrium represents the state of a system in which the properties like pressure, temperature, concentration do not show any change with time.
- Any system which attain equilibrium, two opposite processes are involved.
- Equilibrium is attained when the rate of these opposite processes become equal.

* $H_2O_{(s)} \underset{P-2}{\overset{P-1}{\rightleftharpoons}} H_2O_{(l)}$] → Physical Eq.

Done

* $PCl_{5(g)} \underset{P-2}{\overset{P-1}{\rightleftharpoons}} PCl_{3(g)} + Cl_{2(g)}$] → Chemical Eq.

* If two opposite Process involve physical change :→ Physical Eq.

* _____ Chemical _____ :→ Chemical Eq.

Physical Equilibrium

* Physical Equilibrium :→

Done

(1) Solid $\rightleftharpoons$ liquid.

(2) liquid $\rightleftharpoons$ Vapour.

(3) Solid $\rightleftharpoons$ Vapour.

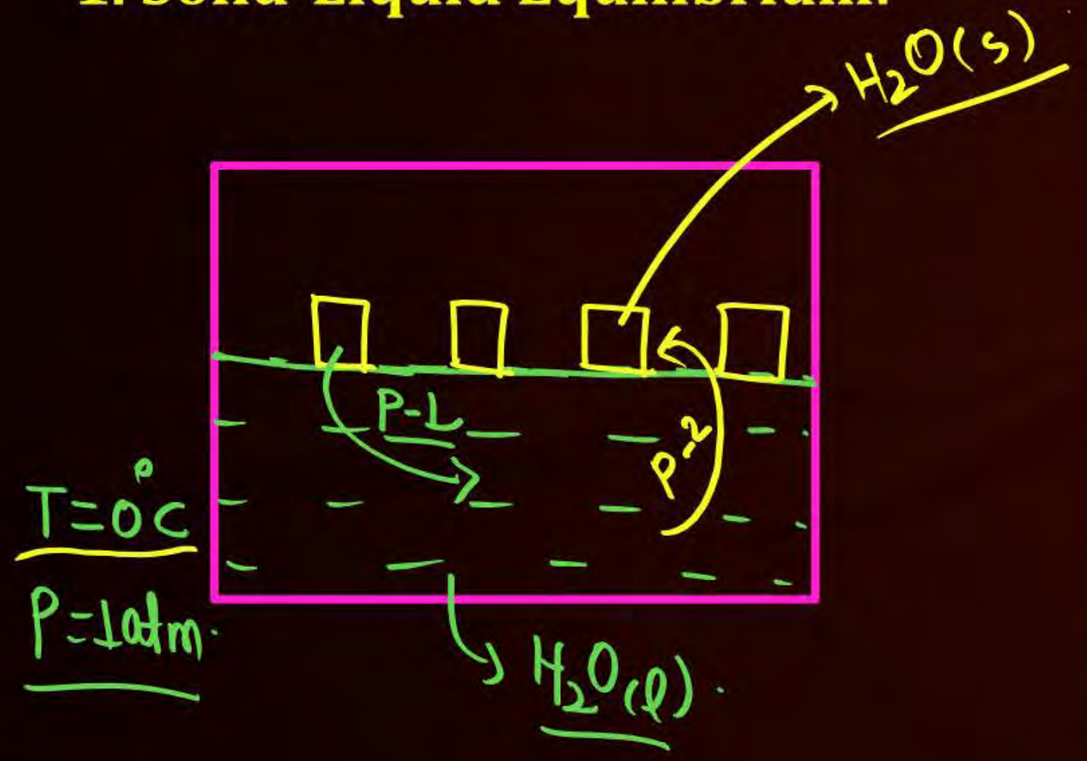
(4) Solid $\rightleftharpoons$ Solution.

(5) Gas $\rightleftharpoons$ Solution.

→ Dissolution of solid and gas in liquid.

Physical Equilibrium

1. Solid-Liquid Equilibrium:



* $H_2O(s) \underset{P-2 \rightarrow \text{freezing}}{\overset{P-1 \rightarrow \text{melting}}{\rightleftharpoons}} H_2O(l)$ } At $0°C$

At eq :→

Rate of melting = Rate of freezing

Done

Physical Equilibrium

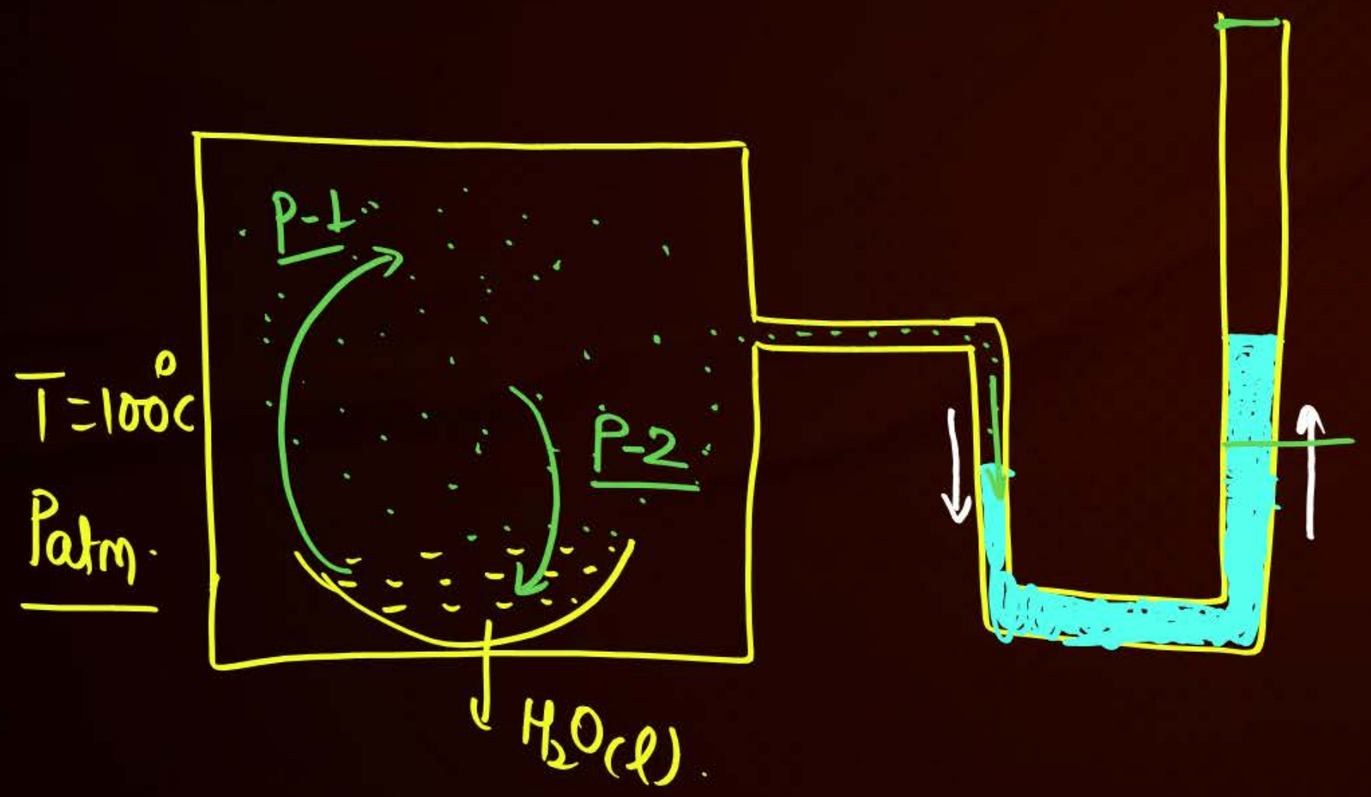
Done

1. Solid - Liquid Equilibrium:

- For any pure [redacted] substance. at one atmospheric pressure (1.013 bar), the temperature at which the solid and liquid phases are at equilibrium, is called normal freezing or melting point of the substance.

Physical Equilibrium

2. Liquid-Vapour Equilibrium:



$$H_2O_{(l)} \underset{P-2}{\overset{P-1}{\rightleftharpoons}} H_2O_{(g)} \rightarrow \text{At } 100^\circ C$$

(B.P.)

At equilibrium :→

Rate of vaporisation = Rate of Condensation

Physical Equilibrium

Done

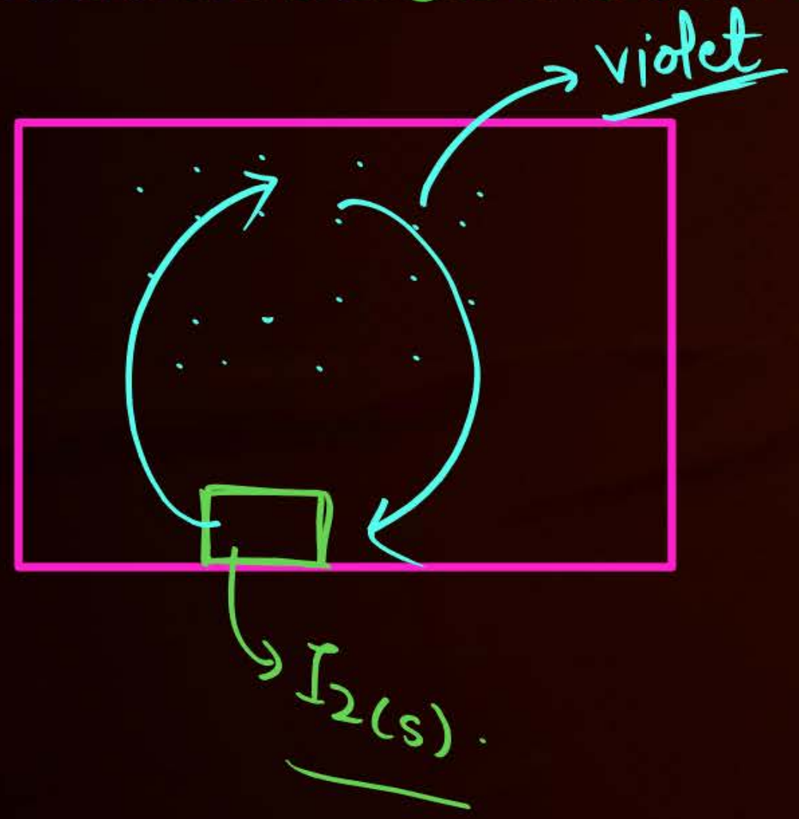
2. Liquid-Vapour Equilibrium:

- For any pure liquid at one atmospheric pressure (1.013 bar), the temperature at which the liquid and vapours are at equilibrium is called normal boiling point of the liquid.
- Boiling point of the liquid depends on the atmospheric pressure.
- It depends on the altitude of the place; at high altitude the boiling point decreases.

Physical Equilibrium

Camphor.

3. Solid – Vapour Equilibrium:



$$I_{2(s)} \underset{P-2}{\overset{P-1}{\rightleftharpoons}} I_{2(vap.)}$$

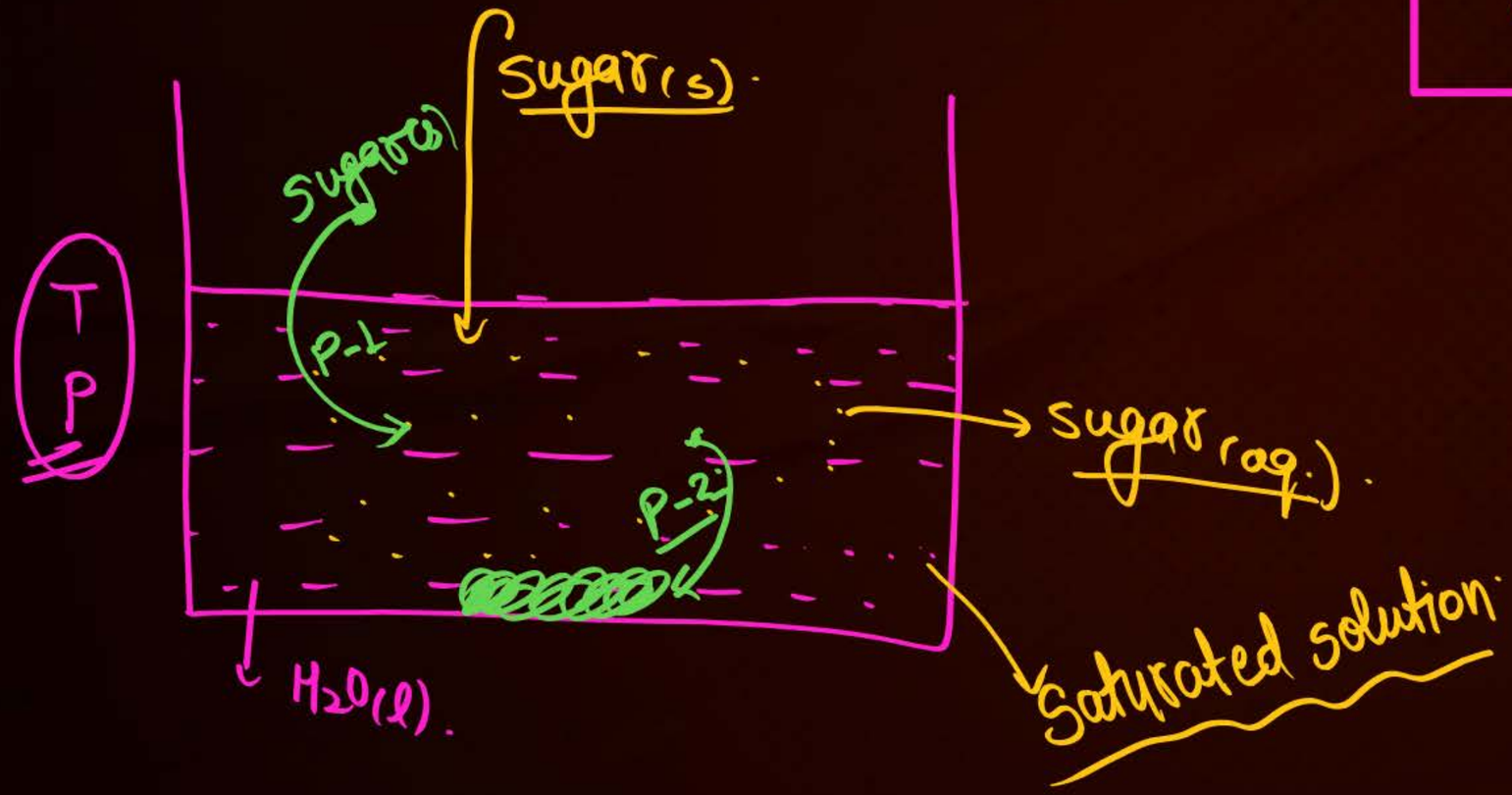
At eq. :→

Rate of sublimation = Rate of deposition.

Physical Equilibrium

Done

4. Dissolution of Solids in liquids:



$$Sugar_{(s)} \underset{P-2}{\overset{P-1}{\rightleftharpoons}} Sugar_{(aq)}$$

At equilibrium :-

Rate of Dissolution = Rate of Crystallisation

↓

Saturated solⁿ

Physical Equilibrium

Solubility:

- The maximum amount of solute that can be dissolved in a specified amount of solvent at a particular temperature and pressure is known as solubility of that solute at that temperature.

Physical Equilibrium

Done

5. Dissolution of Gases in liquids:

Henry's Law: It states that the mass of a gas dissolved in a given mass of a solvent at any temperature is proportional to the pressure of the gas above the solvent.

$$CO_{2(g)} \rightleftharpoons CO_{2(aq.)}$$

Henry's law:

$$m \propto P$$

Pressure of gas.

mass of gas Dissolved in liquid.

$$m = k_H \cdot P$$

Henry's Constant

Physical Equilibrium

Done

General Characteristics of Equilibria Involving Physical Processes:

(i) Possible only in a closed system at a given temperature.

(ii) Both the opposing processes occur at the same rate.

(iii) All measurable properties of the system remain constant.

(iv) When equilibrium is attained for a physical process, it is characterised by constant value of one of its parameters at a given temperature.

Equilibrium in Chemical Process

Chemical Equilibrium

Done

Law of Mass Action:

- Rate of a chemical reaction is directly proportional to the product of active mass of the reactants raised to the power their stoichiometric coefficients.

$1A \longrightarrow$ Product.

Rate of Rxn $\propto [A]$ → Active mass of A

Rate of Rxn $= K[A]$.

↳ Rate constant.

$1A + 1B \longrightarrow$ Product

Rate of Rxn $\propto [A]\cdot[B]$

Rate of Rxn $= K[A][B]$.

↳ Rate constant.

$aA + bB \longrightarrow$ Product.

Rate of Rxn $\propto [A]^a [B]^b$

Rate of Rxn $= K[A]^a [B]^b$

↳ Rate constant.

Chemical Equilibrium

Law of Mass Action:

	Active mass.
* Gas	Molar concentration or Pressure.
* Solute | Ions	Molar Concentration (mol/litre)
* Pure liquid | pure solid | solvent.	→ Active mass is constant and is Considered with the Proportionality constant.

Ex:→ x g. of NaOH is dissolved in V litre of solution.

→ solute.

$[NaOH] = ?$

$n_{NaOH} = \frac{w}{M_w} = \frac{x}{40}$

Ans:→ $[NaOH] = \frac{mol.}{litre}$

$= \frac{x}{40(V)}$

* $C_{(s)} + O_{2(g)} \longrightarrow$ Product

Rate of Rxn $\propto [C][O_2]$

Pure solid (Active mass is constant)

Rate of Rxn $= K[C][O_2]$

Rate of Rxn $= K[O_2]$.

Rate of Rxn $\propto [O_2]$.

$= \frac{mol}{litre} = \frac{n}{V} = \frac{w}{M_w V}$

$= \frac{d}{M_w}$

* $PCl_5(g) \underset{R_b}{\overset{R_f}{\rightleftharpoons}} PCl_3(g) + Cl_2(g)$

* Rate of forward Rx^n (R_f) $\propto [PCl_5]$

$$R_f = k_f [PCl_5]$$

k_f → Rate constant of forward Rx^n.

* Rate of Backward Rx^n (R_b) $\propto [PCl_3][Cl_2]$

$$R_b = k_b [PCl_3][Cl_2]$$

k_b → Rate constant of Backward Rx^n.

Chemical Equilibrium

Consider a chemical reaction:

$$aA(g) + bB(g) \underset{R_b}{\overset{R_f}{\rightleftharpoons}} cC(g) + dD(g)$$

* $R_f \propto [A]^a [B]^b$

$R_f = K_f [A]^a [B]^b$

* $R_b \propto [C]^c [D]^d$

$R_b = K_b [C]^c [D]^d$

* with time :->

$[A], [B] \Rightarrow \downarrow$

$[C], [D] \Rightarrow \uparrow$

* $R_f \downarrow, R_b \uparrow$

At eq :->

$$\boxed{R_f = R_b}$$

↳ Active mass (molar conc.) of both reactants and products becomes constant.

Done

Done

Chemical Equilibrium

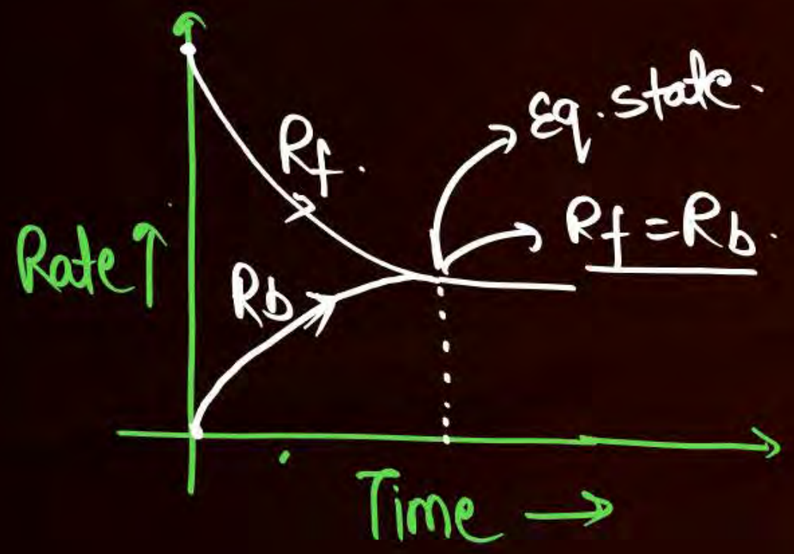
10 mol R ⇌ P

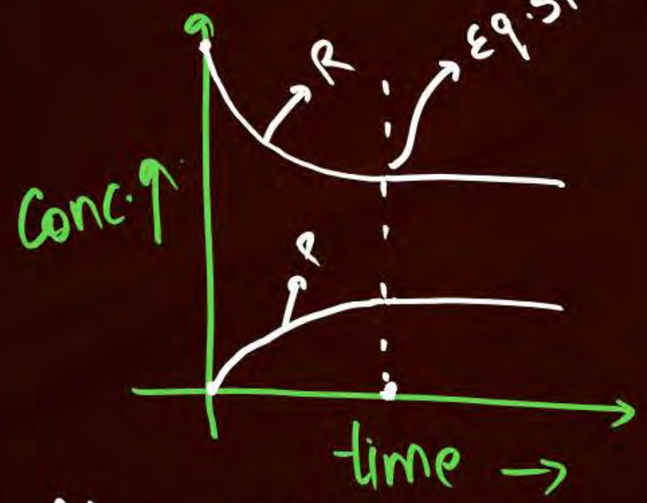
8 mol 2 mol

2 mol 8 mol

5 mol 5 mol

➢ Graphical Representation: [for Reversible Rx^n]





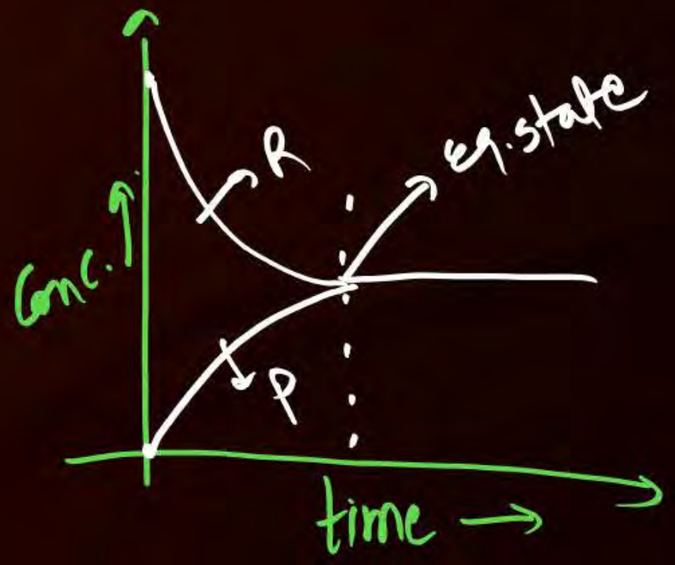
↳ At eq. ∴ [R] > [P]

→ Eq. favours Reactant.

→ Rx^n Hardly Proceed.

↳ At eq. [P] > [R]

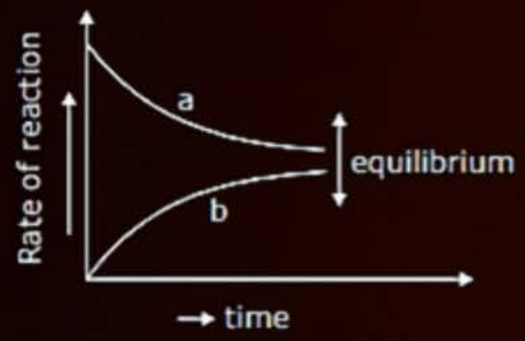
↳ Eq. favours Product.



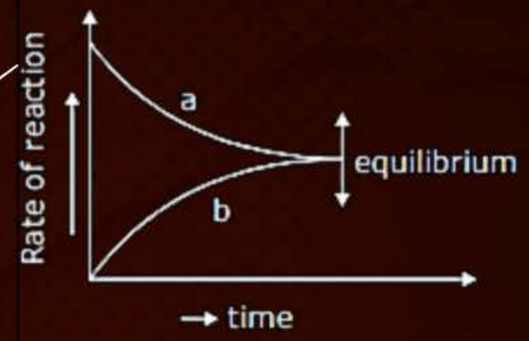
Question

For the equilibrium $A \rightleftharpoons B$, the variation of the rate of the forward (a) and reverse (b) reaction with time is given by (2020 Main, 4 Sep. I)

A



B



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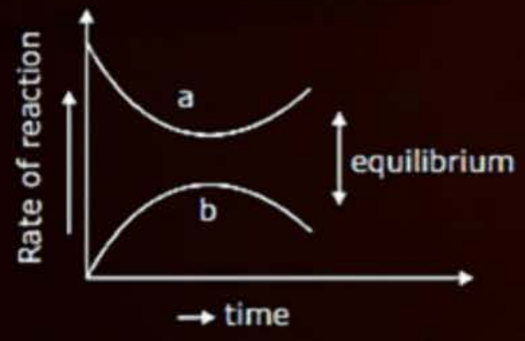
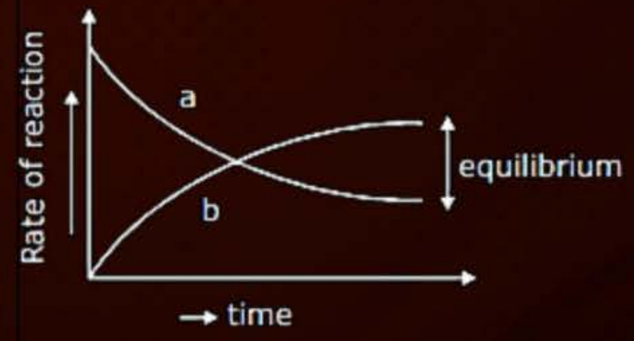
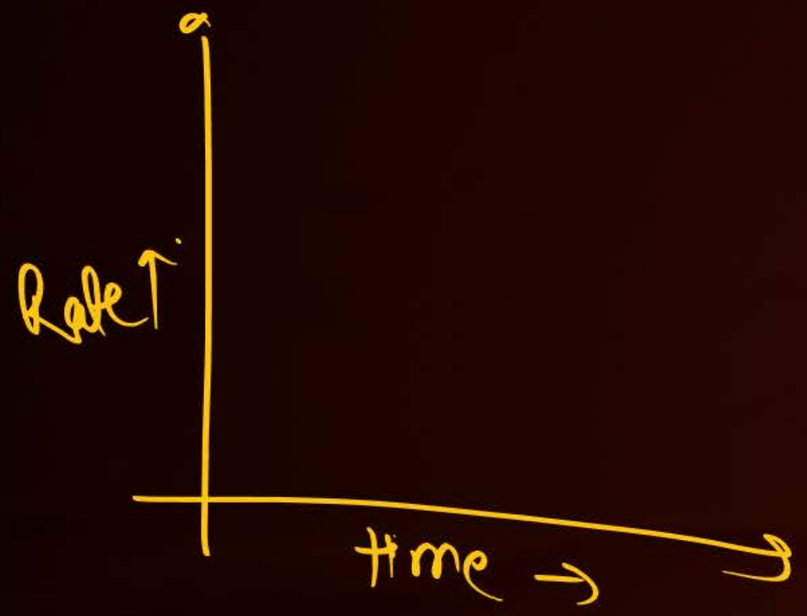


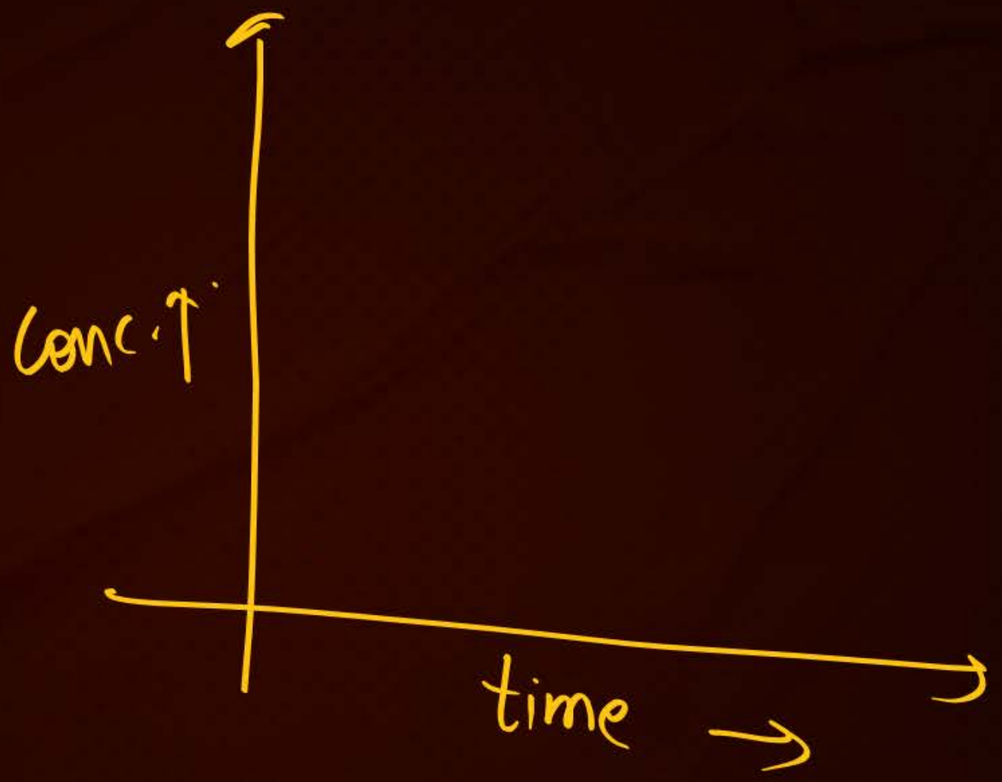
D



HW

* for Irreversible Rx^n :→





HW

Chemical Equilibrium

Characteristics of chemical equilibrium:

- Rate of forward reaction becomes equal to the rate of backward reaction.
- Concentration of all the reactants and products becomes constant.
- Dynamic in nature.
- Equilibrium is attained when none of the product is allowed to escape out or separate out from the reaction mixture.

Chemical Equilibrium

HW

Characteristics of chemical equilibrium:

- Equilibrium can be attained from either directions, from the direction of reactants or products.
- A catalyst does not alter the state of equilibrium. The only effect is that the equilibrium is attained quickly. It increases the rate of both forward and backward reaction to the same extent.

Aaj ke liye
itna hi
Physics Wallah

Homework from Module

- DPP X.
- Revision.

THANK *You*